AWS ADVANCED NETWORKING – STUDY GUIDE

1. VPC & Subnetting

- CIDR, subnetting strategy, IP allocation planning.
- Public, private, isolated subnets.
- Route table design and blackholing detection.

2. Hybrid Connectivity

- AWS Direct Connect architecture, resiliency models (LAG, HA).
- VPN types: Site-to-Site, Accelerated VPN, VPN over DX.
- Routing options: Static, BGP, ASN usage.

3. Transit Gateway (TGW)

- Attachments: VPC, VPN, DX, Connect.
- Route table separation and segmentation.
- Inter-region peering and bandwidth constraints.

4. VPC Peering & PrivateLink

- Peering limitations (no transitive routing, overlapping CIDR).
- PrivateLink architecture, endpoint services, cross-account design.

5. Load Balancing & Global Networking

- ALB, NLB, GWLB use cases.
- Amazon Route 53 routing policies (latency, geolocation, failover).
- Route 53 Resolver inbound/outbound endpoints.

6. Security & Monitoring

- NACL vs Security Groups.
- AWS Network Firewall, WAF, Shield.
- VPC Flow Logs, Traffic Mirroring.

7. IPv6 Networking

- Dual-stack subnets, EUI-64 addressing.
- IPv6 egress-only internet gateway.

8. Advanced Routing & Performance

- Asymmetric routing issues and fixes.
- MTU tuning for DX/VPN.
- Gateway Load Balancer for inline appliances.

9. Troubleshooting

- Reachability Analyzer & Network Access Analyzer.
- Trace route patterns across TGW/VPC.
- BGP status codes and common failures.

10. Exam Tips

- Understand multi-account networks.
- Memorize TGW and DX design constraints.
- Expect scenario questions involving failover and routing design.